Harshit Singh

Data Scientist | Machine Learning • Deep Learning • Computer Vision • Time-Series Forecasting

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Education

Vellore Institute of Technology, Chennai — Chennai, India
*B.Tech, Electronics and Computer Engineering — CGPA: **7.97*** — *2023 – 2027*

New Era School (CBSE) — Ghaziabad, Uttar Pradesh
*Class XII: **87.16%** — Class X: **89.83%*** — *2020 – 2023*

Experience

AI & Robotics Intern — Autonomous Mobile Robotics [**Certificate**] [**GitHub**] — Jul 2025 – Aug 2025
STEMTEC AI & Robotics Technology Pvt. Ltd. — *Chennai, India*

- Quantified positional and orientation drift by running time-series error analysis on **15 Hz** Extended Kalman Filter telemetry against raw encoder logs, measuring a **60% reduction** in mean spatial error (**0.45 m** → **0.18 m**) and **69.2%** in angular error across 10 m ground-truth trials.
- Modeled high-frequency LiDAR telemetry (∼**3,600 points/sec** at **10 Hz**) to validate a windowed spatial-filtering algorithm, achieving **95% obstacle-detection accuracy** across 20 randomized stress-test trials with **sub-50 ms** end-to-end telemetry latency.

Projects

SPARC — Sign Language Recognition *(Computer Vision, Random Forest)* [**Demo**] [**Web-App**]

- Engineered a **229-dim** geometric feature space (joint angles + pairwise distances) from **921,600-dim** raw input, enabling **92.4% accuracy** at 15–20 FPS via Random Forest.
- Cut false-positive errors by **92%** across **41** gesture classes via sliding-window filtering — **Top 5 national finalist** (500+ teams), Smart India Hackathon 2025. [**Certificate**]

DrishtiGuide — Edge Vision & Assistive AI *(YOLOv8, Sensor Fusion)* [**GitHub**]

- Built a real-time edge CV pipeline fusing YOLOv8 detection with monocular distance estimation ($F = 700$**px**), cutting inference overhead by **66.7%** at **sub-100ms** latency.
- Designed a 6-DOF IMU fall-detection algorithm and LBPH facial recognition system, achieving >**95%** telemetry reliability with zero false positives during normal gait.

Weather Risk AI — Forecasting & Graph Risk Engine *(LSTM, NetworkX)* [**GitHub**]

- Architected a 2-layer stacked LSTM model across **6** meteorological variables from **1,460+** daily records via 10-day sliding windows, achieving **sub-0.05 MSE**.
- Modeled cascading flood risk across **8** zones via directed-graph diffusion (**0.8** decay, 5 iterations), computing drainage limits via Edmonds-Karp max-flow and Dijkstra's algorithm.

NeuroBridge AI — Clinical Analytics Platform *(MediaPipe, Multimodal ML)* [**Live**] [**GitHub**]

- Architected a multimodal late-fusion risk model (0.40·**Vision** + 0.30·**Behavioral** + 0.30·**Engagement**) combining 468-landmark facial biomarkers with clinical survey data.
- Integrated a TensorFlow.js **NLP** toxicity classifier (**7** categories, $\tau = 0.85$ confidence) for content moderation, alongside OLS-based 30-day patient risk forecasting.

Skills

Languages: Python, SQL, C, Java
Machine Learning: Scikit-Learn, TensorFlow/Keras, Random Forest, LSTM, Time-Series Forecasting, Feature Engineering
Computer Vision: OpenCV, YOLOv8, MediaPipe, TensorFlow.js
Data Tools: Pandas, NumPy, Matplotlib, SQLite, NetworkX, Git, Cloud Deployment (Render)
Professional Skills: Data-Driven Problem Solving, Cross-Functional Collaboration, Communication, Time Management

Achievements & Leadership

- **3rd Place, Technovation 2025** (Kurukshetra '25, CEG Guindy, Anna University) — ranked **2nd Runner Up out of 50+** teams; **INR 15,000** award. [**Certificate**]
- **Winner, Innovate X Impact (V-NEST)** — secured **1st Position out of 35** teams; **INR 5,000** award.
- **Deputy Operations Lead, Team Namo Nirvana (VIT Chennai)** — led **100+ member** teams across **5–6 competitions/semester**, boosting efficiency by **35%** and securing **2 sponsors.** [**LOA**] [**Certificate**] [**LOR**]

Languages

English (Fluent) | **Hindi** (Native)